Calculation Workbook Queries

Questions arising from the six calculation sheets, and what each one blocks

To: Vitech Enviro Systems Pvt. Ltd. (Engineering / Applications Team)
From: [Your Company Name]
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Why this document exists

You supplied six calculation workbooks. We have transcribed every formula in them and implemented four - the VOC and LEL check, the tank and oven heat loads, the scrubber and duct diameters, and the structural stock weights - each verified against the worked example on your own sheet.

The eight questions below are what implementing them raised. They are not requests for more documents. Each one is a point where your sheets either disagree with each other, or where a cell does something its own description does not, and where guessing would put a number we invented into a document a customer reads.

> *Answer Q1 first if you answer nothing else. It decides the airflow, and the airflow decides the blower, the motor, the filters, the ducting and the price of every paint booth the platform quotes.*

The eight questions at a glance

#	Area	Question	Blocks
Q1	Blocks Every Booth	Which dimension is the extracted face, and which height?	Airflow, blower, filters, ducting and price of every paint booth
Q2	Safety-Adjacent	Dry-off oven air density - 101.325 or 1.204?	Dry-off oven heater sizing
Q3	Confirm A Typo	The oven steel-mass cell adds an area to a mass	Both oven heat loads
Q4	Blocks Scrubber Sizing	Scrubber and duct diameters - what is the standard ladder?	Scrubber and duct sizing, and the GA drawing that follows it
Q5	Costing	MS flat - 12 kg or 16 kg per 6 m length?	Structural weight in the BOM and the cost
Q6	Costing	Painting rate - Rs 35 or Rs 50 per sq.ft?	Rate card and every painted line in a quotation
Q7	Costing	An unlabelled Rs 70,000 line	Cyclone / cartridge cost model
Q8	Commercial Policy	How is the margin multiplier chosen?	The whole quotation margin model

1. The questions

Q1. Which dimension is the extracted face, and which height?

BLOCKS EVERY BOOTH

What we found

- In `Standard Booth.xlsx` the airflow formula is =1.5*2.4*0.5*3600, where 1.5 is the L column and 2.4 is H. The W column is a menu of options (1250 / 1500 / 2250 / 3000) and never enters the calculation at all.
- The costing workbook agrees: cells K5 and L5 take L and H, giving 3.0 x 2.4 = 7.2 m2 and 12,960 CMH for the 3.0m L x 2.4m W x 2.4m H booth.
- The AI knowledge-base PDF you supplied uses the OPPOSITE naming - it calls that same open front W and the depth D - and computes on a 1.5 m EFFECTIVE FILTER OPENING, not the full height: CMH = W x 1.5 x 0.5 x 3600.

Why it matters

For one 3.0 m wide booth the two documents give 12,960 CMH and 8,100 CMH - a 60% difference. That selects a different blower, a different motor, a different filter count and a different price.

What we need from you

- For a booth you describe as 3.0m L x 2.25m W x 2.4m H, is the extracted face 3.0 x 2.4, or 3.0 x 1.5?
- When a customer states three dimensions in an enquiry, which one is the open front? We currently read the SECOND number as the face width, which is your depth.

Q2. Dry-off oven air density - 101.325 or 1.204?

SAFETY-ADJACENT

What we found

- `Heat Load.xlsx` -> Dry off Oven, cell N13 gives the air density as 101.325. That is standard atmospheric PRESSURE in kPa, not a density.
- The Curing Oven sheet uses 1.204 kg/m3 for the same quantity, which is the density of air.
- Read as a density, 101.325 is about 84x too high. On your worked oven the air term becomes 86,670 of the 188,786 Kcal total - nearly half the heater.

Why it matters

If 101.325 is a typo, every dry-off oven sized from this sheet is substantially oversized. If it is deliberate (a leakage or air-change allowance folded into one number), we need to know that, because we would otherwise 'correct' a figure you intended.

What we need from you

- Should the dry-off oven use 1.204 kg/m3, or is 101.325 carrying something else?

Q3. The oven steel-mass cell adds an area to a mass

CONFIRM A TYPO

What we found

- `Heat Load.xlsx` -> Dry off Oven D18 (and Curing Oven D17, identically):
- =ROUNDUP(((D8/1000*F8/1000)*2)+((E8/1000*F8/1000)*2)+((D8/1000*E8/1000)*3)*N12/1000*H18,0)
- The '* density * thickness' at the end binds to the THIRD term only. So the two wall terms - 25.8 and 16.5 square METRES on your worked oven - are added straight to a mass in kilograms.
- The cell returns 377 kg. The formula as written in words gives 733 kg. Your curing oven reads 1,647 kg where the same wording gives 3,016 kg.

Why it matters

We reproduce your cell exactly, so your totals (188,786 Kcal / 220 kW) come out right - but we cannot adopt a formula that adds m2 to kg. Until this is confirmed our oven figures will differ from yours by exactly this amount.

What we need from you

- Is the bracket a typo? If so we will use the sound reading and your sheets should be corrected, since both oven sheets carry it.

Q4. Scrubber and duct diameters - what is the standard ladder?

BLOCKS SCRUBBER SIZING

What we found

- `Vertical Scrubber - Diameter calculation.xlsx` computes D = 1545 mm at 6750 CMH (cell D12), and the row beneath it (D13) reads '~ 950'.
- 950 mm corresponds to roughly 2550 CMH, not 6750. The same happens on both ducts: computed 399 mm, typed '~ 300' and '~ 350'.
- We checked the cells: those three rows are hand-typed TEXT, not formulas, so they appear to be leftovers from an earlier run.

Why it matters

We can compute the diameter exactly as your sheet does, and we do. We cannot round it to a size you actually build without knowing your standard diameters and whether you round up or to nearest.

What we need from you

- What are your standard tower and duct diameters?
- Do you round up to the next standard size, or to the nearest?

Q5. MS flat - 12 kg or 16 kg per 6 m length?

COSTING

What we found

- The paint-booth sheet costs MS Flat 40 x 6 x 6000 at 12 kg per length.
- The cyclone sheet costs MS FLAT 40x6x6000 at 16 kg per length.
- 40 x 6 mm x 6 m of mild steel weighs about 11.3 kg, so 12 matches the steel and 16 does not.
- Note we resolved the square-tube difference ourselves - the booth uses 40x40x3 (21 kg) and the cyclone 40x40x2 (18 kg), two different sections. Only the flat remains.

Why it matters

Structural weight feeds the BOM and the cost. A 33% error on a section runs straight into the quoted price.

What we need from you

- Which figure governs for MS flat, and is the 16 kg row a different section?

Q6. Painting rate - Rs 35 or Rs 50 per sq.ft?

COSTING

What we found

- Within `Cyclone recovery & Cartridge filter unit.xlsx`, the cartridge filter unit costs painting at Rs 35/sq.ft (cell H14) while the cyclone and the ducting use Rs 50/sq.ft (H14 and H23 of the other sheet).
- The paint-booth sheet uses Rs 35/sq.ft.

Why it matters

On the cartridge unit alone the painting line is Rs 75,250 - it is not a rounding difference.

What we need from you

- Is Rs 50 a different paint specification (two-coat, epoxy, a different surface preparation), or a rate change?
- Which rate should the platform quote by default?

Q7. An unlabelled Rs 70,000 line

COSTING

What we found

- `Cyclone recovery & Cartridge filter unit.xlsx` -> Combine, cell B5 is a hardcoded 70,000 with no description and no formula. It is 13% of the Rs 5,28,790 works cost.

Why it matters

We cannot reproduce your total without knowing what it buys, and we will not invent a line item to make a total balance.

What we need from you

- What is the Rs 70,000 - blower, panel, erection, something else?

Q8. How is the margin multiplier chosen?

COMMERCIAL POLICY

What we found

- The booth workbook applies x1.40 to the booth and x1.26 to the exhaust duct, adds fixed lines for design (25,000), packing (11,000) and E&C (65,000), then deducts a 10% discount, landing at 27% profit.
- The cyclone workbook instead shows three alternatives - x1.35, x1.25 and x1.17 - beside a typed Rs 7,60,000 and a Rs 1,50,000 deduction, which reads like a negotiation rather than a rule.

Why it matters

Our current model applies a flat 15% allowance for bought-out items, which is why our cost-plus figure diverges from your history by up to 57%. Your per-line multipliers would replace it - we just need the selection rule.

What we need from you

- What decides the multiplier - equipment type, order value, customer, competition?
- Are the fixed lines (design / packing / E&C) always those amounts, or do they scale?
- Is the 10% discount a standard concession or was it specific to that enquiry?

2. What we already settled from the workbooks

So the list above is read for what it is - the genuine remainder, not a request to re-explain your own documents. Everything here was answered by the sheets themselves and needs nothing further from you.

Item	What was in question	Resolution
Face velocity	0.5 m/s, stated in two of your workbooks	Adopted. It replaces the NFPA 33 default of 0.45 we had used while your earlier document was silent. Every booth airflow rose about 11%.
Square tube weight	21 kg vs 18 kg per 6 m	Not a conflict - two thicknesses (40x40x3 and 40x40x2). Both now on file.
The Rs 80,730 gap	Your cost sheet's first row was cropped in the earlier image	Recovered: MS 18 SWG sheet, 621 kg, RM 52,785 + labour 27,945. Your total of Rs 6,49,264 now reconciles exactly.
Booth sheet weight	We computed 1,240 kg, our pricing model assumed 3,645 kg	Your panel module settles it: 27 panels x 23 kg = 621 kg. Both of our figures were wrong.
Rates	Rs 85 + 45 /kg sheet, Rs 75 + 50 /kg sections, Rs 3,500/HP	All confirmed against your cost sheet, and the cyclone sheet's Rs 130/kg is the same 85 + 45.
Blower selection	9000 CFM booth duty	Our selector reproduces your BOM line CLP-4-10-9000 exactly.

3. What happens when you answer

Q1 releases the booth engine: the airflow, blower, filter and duct chain, the panel-count weight model and a validated booth cost model that reconciles to your own Rs 6,49,264. Q2 and Q3 release the oven heat loads. Q4 releases scrubber sizing. Q5 to Q8 release the costed BOM and the quotation margin.

One further note, offered rather than asked. Your AI database PDF lists a real standard range - VT/1.5 through VT/4.5, front-open and enclosed, wet and dry, each with its airflow and motor. The platform currently engineers every booth from first principles. Once Q1 is settled it could instead recognise a standard model and quote it directly, which is both faster and closer to how you actually sell.

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